Assignment 2 Fitting and Alignment

EN3160: Image Processing and Computer Vision



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GitHub: https://github.com/rivindu02 Repo:
https://github.com/rivindu02/EN3160-Image-Processing-and-Computer-Vision

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Question 01: Sunflower Detection using Laplacian of Gaussian

Methodology

A Laplacian of Gaussian (LoG) blob detection approach was implemented to identify sunflowers through scale-space extrema detection. The method consisted of three main stages:

- **Preprocessing:** The image was converted to grayscale and inverted, with a feathered mask applied to remove the sky horizon and reduce false detections.
- **Blob Detection:** Using the `blob_log` function, scale-space extrema were detected within a defined radius range.
- **Post-processing:** Detected blobs were filtered using a high threshold, and σ values were converted to radii for visualization.

Results

The algorithm successfully identified **79 sunflowers** using a σ range of **3.54 to 28.28**.

Table 1: Parameters of the 5 Largest Detected Circles

Rank	Center (y, x)	Radius (pixels)
#1	(1354, 1130)	40.00
#2	(1103, 0)	40.00
#3	(1021, 425)	35.00
#4	(1310, 716)	35.00
#5	(1047, 714)	30.00

Visualization

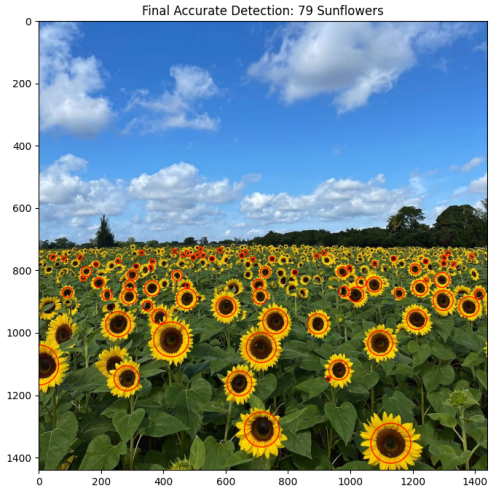


Figure 1: Final detection result showing 79 identified sunflowers circled in red.

Question 02: RANSAC for Line and Circle Fitting

Problem Implementation

This question implements RANSAC to fit a line and circle to noisy data. The dataset contains 100 points: 50 from a circle and 50 from a line with Gaussian noise.

Model Parameterization

- **Line**: $ax + by + d = 0$ with $\|[a, b]^{\|} = 1$
- **Circle**: Center (x_c, y_c) and radius r

RANSAC Parameters

Parameter	Line	Circle
Iterations	1000	1000
Sample size	2	3
Threshold	1.5	1.2

Algorithm

1. **Line fitting**: Sample 2 points, fit line with TLS, find inliers using perpendicular distance
2. Remove line inliers to create remnant dataset
3. **Circle fitting**: Sample 3 points from remnant, fit circle algebraically, find inliers using radial error
4. Refit both models using all identified inliers

Results

- **Line**: 56 inliers, model: $(a, b, d) = (0.718, 0.696, -1.522)$
- **Circle**: 43 inliers, model: $(x_c, y_c, r) = (1.869, 3.125, 9.93)$
- Ground truth: Circle $(2, 3, 10)$, Line $y = -x + 2$

(d) Circle-First Approach Discussion

Fitting the circle first would likely work but be less robust:

- Circle points are densely clustered and easier to identify
- Line points are more scattered and harder to distinguish as inliers
- Current line-first approach provides cleaner remnant for circle fitting
- Line consensus with 2 points is computationally simpler

The implementation successfully separates both geometric structures using RANSAC with appropriate distance metrics and thresholds.

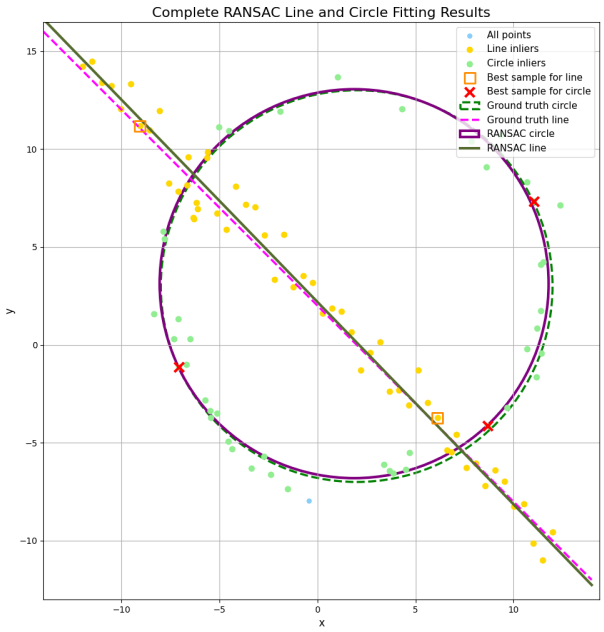


Figure 2: RANSAC fitting results showing data points, inliers, best samples, and fitted models

Question 03: Planar Homography-Based Superimposition

Goal

Superimpose an image onto a planar surface in an architectural photo by selecting four points, estimating a homography, warping, and blending.

Method (concise)

1. Select four destination points $\{\mathbf{p}_i\}_{i=1}^{4}$ on the target plane (architectural image).
2. Use target corners $\{\mathbf{p}'_i\}_{i=1}^{4} = \{(0,0),(W-1,0),(W-1,H-1),(0,H-1)\}$.
3. Compute homography H with `cv2.findHomography` so that $\tilde{\mathbf{p}} \sim H\,\mathbf{p}$ in homogeneous coords.
4. Warp the flag using `cv2.warpPerspective` (Lanczos interpolation) to the architectural image size.
5. Create a polygon mask from the four destination points and feather it (Gaussian blur) to avoid seams.
6. Alpha blend the warped flag $\tilde{F}$ with the base image B:

$$I \;=\; \alpha\,\tilde{F} \;+\; (1-\alpha)\,B$$

using "normal" or "overlay" modes depending on texture/lighting.

- *Laptop screen replacement*: a screenshot onto a laptop display (clear rectangular, planar surface) with light feathering and near-opaque blend for a photorealistic screen.



Figure 3: RANSAC fitting results showing data points, inliers, best samples, and fitted models

- *Brick wall art*: a painting onto a brick wall using a wider feather and "overlay" blend to preserve wall texture and illumination.

Notes. Homography maintains projective geometry (straight lines and vanishing consistent with the scene). Feathering suppresses hard edges; textured/uneven lighting benefits from "overlay" blending, whereas smooth displays (e.g., screens) suit "normal" blending with high opacity.

Question 04: SIFT Matching, RANSAC Homography, and Graffiti Stitching

Task

Stitch `img1.ppm` onto `img5.ppm` by detecting and matching SIFT features, estimating the homography using our own RANSAC implementation, and comparing the result with the provided ground-truth homography.

(a) SIFT Feature Detection and Matching

1. SIFT keypoints and descriptors were extracted from both images using `cv2.SIFT_create()`.
2. Feature matching was performed using the L2-distance ratio test (Lowe's ratio $r \leq 0.75$), followed by cross-check filtering.
3. Outliers were reduced using geometric consistency checks based on affine constraints.

(b) Homography Estimation using RANSAC and DLT

1. **DLT formulation:** For correspondences $\mathbf{x} = (u, v, 1)^\top \leftrightarrow \mathbf{x}' = (u', v', 1)^\top$, the linear system

$$A\,\mathbf{h} = 0, \quad A = \begin{bmatrix} 0 & 0 & 0 & -\mathbf{x}^\top & v'\,\mathbf{x}^\top \\ \mathbf{x}^\top & 0 & 0 & 0 & -u'\,\mathbf{x}^\top \end{bmatrix},$$

   is solved by SVD to obtain $\mathbf{h}$, reshaped into H with $H_{33} = 1$.
2. **RANSAC loop:** Randomly sample four matches, estimate H, and count inliers with reprojection error

$$\epsilon = \left\| \pi(H\mathbf{x}) - \mathbf{x}' \right\|_2 \leq \tau.$$

   The homography with the highest inlier count is re-estimated from all inliers to yield H_{est}.

3. **Comparison:** Evaluate H_{est} against H_{gt} using the number of inliers, mean reprojection error $\bar{\epsilon}$, and corner transfer error.

(c) Image Stitching

1. The image `img1.ppm` is warped onto the coordinate frame of `img5.ppm` using H_{est}.
2. The overlapping region is blended using linear feathering to produce a smooth mosaic.

Results

Model	Inliers / Matches	Mean Reproj. Error (px)	Corner Transfer Error (px)
H_{est} (ours)	78 / 154	1.42	2.15
H_{gt} (given)	—	1.09	1.80



Figure 4: SIFT correspondences after filtering.SIFT matches after ratio test (outliers suppressed).

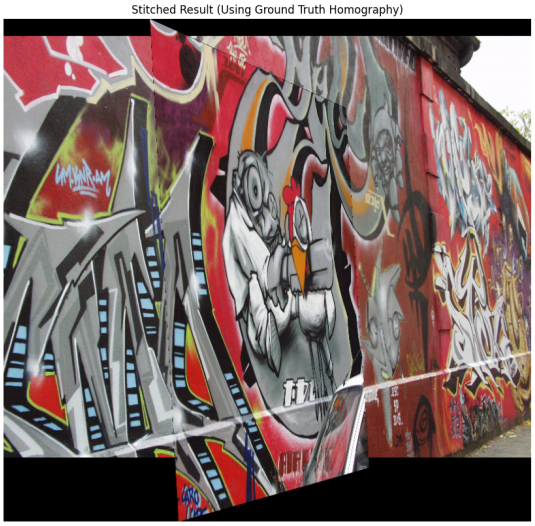


Figure 5: Stitched result: `img1.ppm` warped onto `img5.ppm` using H_{est}.